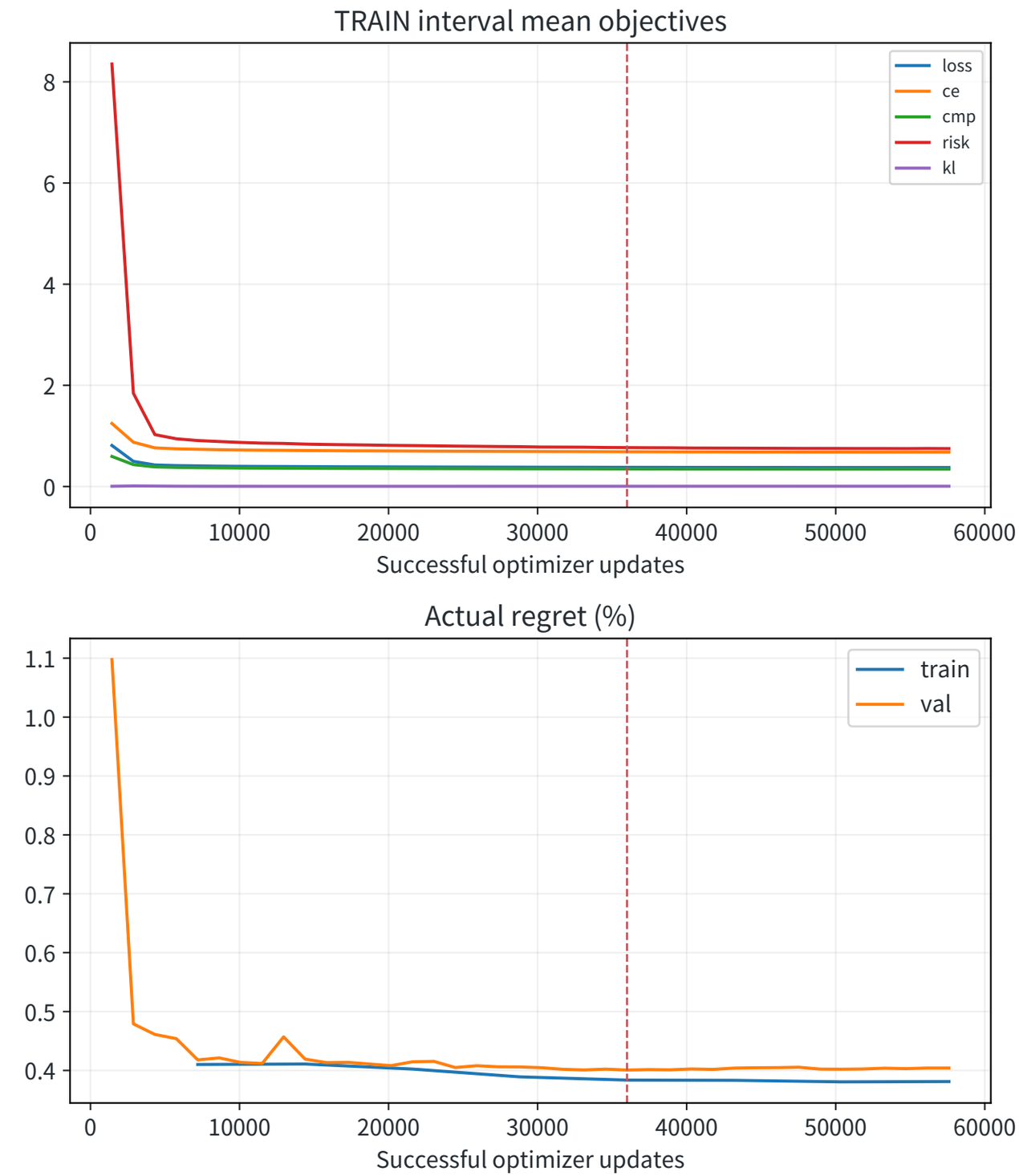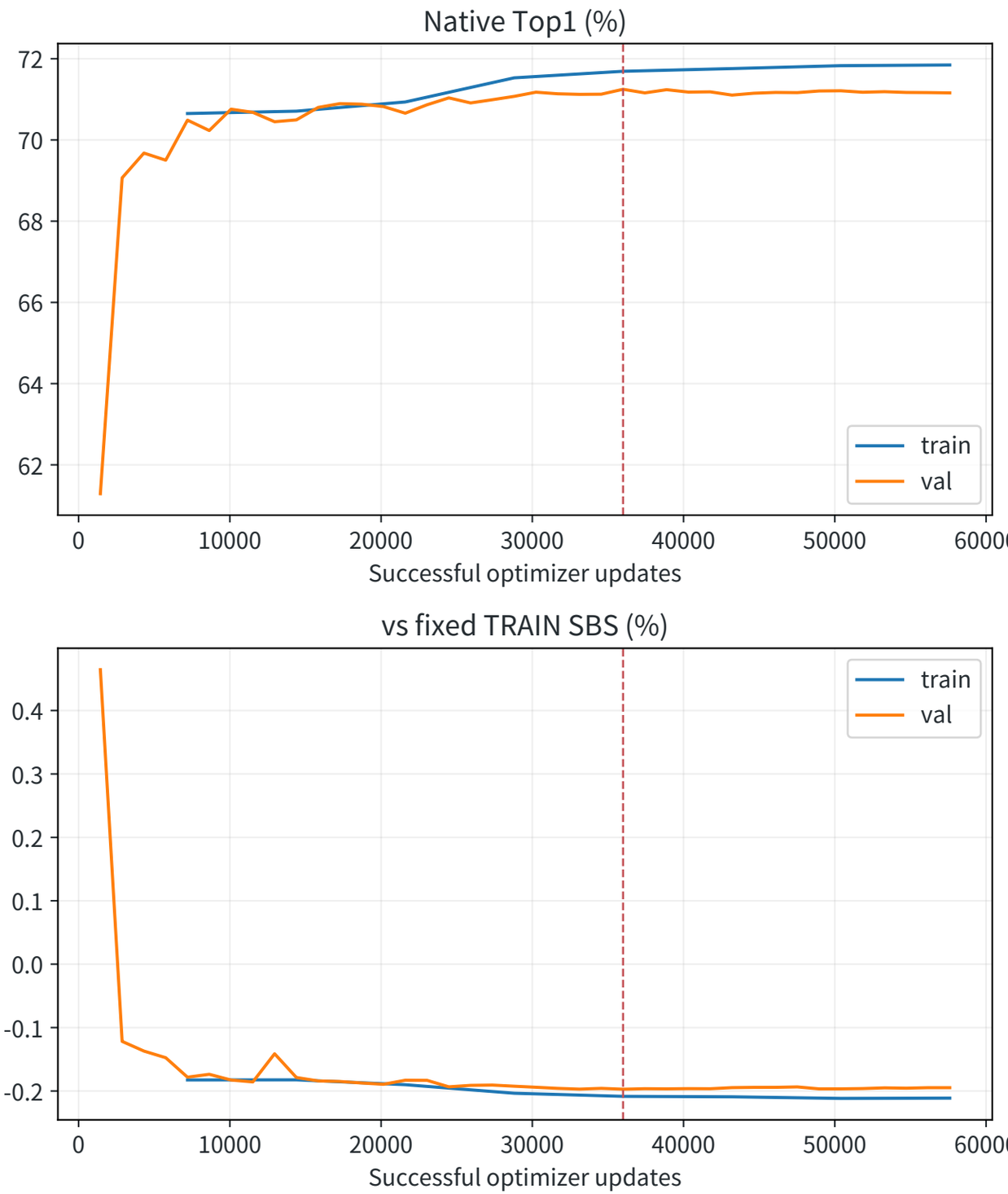R89/B_muon_hybrid | successful updates; one locked VAL checkpoint